Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 24010101116

Lab - 11

Roll No : 366

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:

Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

In [5]:
```
li = []
fp = open("helloworld.txt","r")
ch = len(fp.read())
fp.seek(0,0)
rl = fp.readlines()
for i in rl:
    for j in i.split(" "):
        li.append(j)
fp.close()
fp = open("output1.txt","w")
fp.write(f"Total No. Of Lines: {len(rl)}.\nTotal No. Of Words: {len(li)}.\nTotal No
fp.close()
```

Practical 2: Search a Word in a File

Problem Definition:

Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

In [11]:
```
li1 = []
fp = open("helloworld.txt","r")
rli = fp.readlines()
for i in rli:
    for j in i.split(" "):
        li1.append(j)
fp.close()
w = input("Enter a word: ")
fp = open("output2.txt","w")
fp.write(f"Occurrences Of the word {w} in the given file is {li1.count(w)}.")
fp.close()
```

```
Enter a word: is
```

Practical 3: Student Marks and Average

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

In [45]:
```
def stuAvg(li):
    sum = 0
    l = len(li)
    for i in range(2,l):
        sum += int(li[i])
    return sum/(l-2)
sli = []
fp = open("student1.txt","r")
rel = fp.readlines()
for i in rel:
    sli.append(i.split(","))
fp.close()
fp = open("output3.txt","w")
for i in sli:
    avg = stuAvg(i)
    fp.write(f"Student with rollNo. {i[0]} and Name {i[1]} got average marks: {avg}
fp.close()
```

Practical 4: Maximum and Minimum using Tuple

Problem Definition:

Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

In [30]:
```
t = tuple()
fp = open("numbers.txt","r")
r = fp.read()
for i in r.split(","):
    t += (int(i),)
fp.close()
fp = open("output4.txt","w")
fp.write(f"Maximum value: {max(t)}\nMinimum value: {min(t)}")
fp.close()
```

Practical 5: Remove Duplicate Words using Set

Problem Definition:

Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

In [36]:
```
lis = []
fp = open("helloworld.txt","r")
r = fp.readlines()
for i in r:
    for j in i.split(" "):
        lis.append(j)
fp.close()
s = set(lis)
fp = open("output5.txt","w")
for i in s:
    fp.write(f"{i}\n")
fp.close()
```

Practical 6: Word Frequency using Dictionary

Problem Definition:

Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

In [48]:
```
li6 = []
d = {}
fp = open("helloworld.txt","r")
readl = fp.readlines()
for i in readl:
    for j in i.split(" "):
        li6.append(j)
for i in li6:
    if i not in d.keys():
        d[i] = li6.count(i)
fp.close()
fp = open("output6.txt","w")
for k,v in d.items():
    fp.write(f"{k}: {v} times.\n")
fp.close()
```

Practical 7: Student Result and Grade Calculation

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C
- Marks < 40 : Fail

In [44]:
```
def grade(m):
    if m >= 75:
        return 'A'
    elif m >= 60:
        return 'B'
    elif m >= 40:
        return 'C'
    else:
        return 'Fail'
sli2 = []
fp = open("student2.txt","r")
rel2 = fp.readlines()
for i in rel2:
    sli2.append(i.split(","))
fp.close()
fp = open("output7.txt","w")
for i in sli2:
    g = grade(int(i[2]))
    fp.write(f"Student with rollNo. {i[0]} and Name {i[1]} got {g} Grade.\n")
fp.close()
```

In [ ]: